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The Forrester Wave™: Business Rules Platforms, Q2 2008

by John R. Rymer and Mike Gualtieri

for Application Development & Program Management Professionals

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ILOG, Fair Isaac, Pegasystems, Corticon, And Haley Lead, But Innovations Software Technology Is Also Very Strong

by John R. Rymer and Mike Gualtieri

with Matthew Brown and Catherine Salzinger

EXECUTIVE SUMMARY

Forrester evaluated 13 platforms from 11 vendors using 175 criteria and found ILOG JRules, Fair Isaac Blaze Advisor, and Pegasystems PegaRULES to be resounding Leaders as general-purpose business rules platforms. Haley Limited (Haley Expert Rules), Corticon Technologies, and Innovations Software Technology also emerged as Leaders, and InRule Technology followed close behind. But choosing a business rules platform is not as easy as simply picking one of the Leaders. To help application development pros understand these choices, Forrester created five views of the business rules market, including: general-purpose platforms, specialized platforms (from Experian, Intelligent Results, and Haley Limited's Haley Office Rules and Determinations Engine), platforms for Java application developers, platforms for .NET application developers, and platforms for business analysts who develop and maintain applications.

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NOTES & RESOURCES

Forrester conducted questionnaire-based product and strategy research during November and December 2007 and interviewed 11 vendors: CA, Corticon Technologies, Experian, Fair Isaac, Haley Limited, ILOG, Innovations Software Technology, InRule Technology, Intelligent Results, Pegasystems, and SAP. In addition, Forrester interviewed two reference customers for each vendor's product (26 user companies).

Related Research Documents

"Trends: Business Rules Platforms 2008"
December 21, 2007

"Business Rules And SOA Ease The Burden of Regulatory Compliance"
November 15, 2007

BUSINESS RULES PLATFORMS ARE KEY TO DYNAMIC BUSINESS APPLICATIONS

Rules are an indispensable underpinning of any business — policies and operational procedures are used to make decisions, determine the flow of work, establish prices, make operational choices, enforce constraints, and comply with regulations.¹ And many rules of business change fast. Among organizations using business rules platforms, we found, for example, that the California Board of Realtors' rules governing forms required for real estate sales change every six months. Further, a large electronics distributor must adjust to multilevel rebate programs that change monthly. And in some online retailing scenarios, business rules change every day.

Forrester has identified business rules platforms as a key enabling technology for Dynamic Business Applications because they enable applications to be built for change.² Business rules platforms allow:

- **Companies to manage their business rules independently of applications.** Business rules implemented in Java, COBOL, C#, Visual Basic, C++, PHP, and other languages are so tangled up with other forms of program logic that they are difficult to identify, change, and test. Business rules platforms allow application development teams to “externalize” these rules to ease maintenance.
- **Businesspeople to author and maintain business rules.** Some business rules platforms provide tools for businesspeople — pricing analysts, risk managers, compliance analysts, and marketing and promotions professionals — to create and maintain business rules. For many IT managers, this is a scary notion because of the risk that businesspeople will inadvertently create errors and outages. However, mature application development shops find that given the right tools — and testing, release, and change management processes — they can delegate control over fast-changing rules to business analysts.
- **IT to create rules services for SOA environments.** Business rules platforms now allow application development pros to expose rules as services that are available to multiple applications. This connection between business rules platforms and service-oriented architecture (SOA) makes the goal of common rule sets that can be applied to many applications a reality.³

The Business Rules Market Is A Sprawling Landscape

Business rules platforms are a leading category within a sprawling landscape of rules technologies. The *platforms* are designed for custom application development and provide a full range of programming, configuration, runtime, and administration and management tools. Many of the platforms integrate well with popular Java application servers and/or Microsoft's .NET platforms.

Customers can adopt business rules technology from several other sources in addition to platform vendors.⁴ The other major sources are:

- **Specialized applications.** Clients can purchase business rules technology embedded in applications for industry-specific issues like anti-money-laundering and credit origination. The largest grouping of these applications seems to be in financial services, including solutions from CGI, Experian, Fair Isaac, FINEOS, First Data (which during 2007 purchased Intelligent Results), MindBox, NaviSys, and Pegasystems. In some of the packaged solutions from Experian, Fair Isaac, and First Data and in all of Pegasystems' solutions frameworks, the vendor provides both applications code and an underlying business rules platform.
- **Enterprise applications.** Enterprise applications vendors like Oracle and SAP incorporate business rules into some of their enterprise applications. For example, Oracle Siebel Customer Relationship Management 8.0 incorporates Haley Expert Rules, and SAP has partnerships with Haley Limited and ILOG.
- **BPM platforms.** Most of the business process management (BPM) platforms incorporate business rules in some form. For example, Microsoft provides rules engines for both BizTalk Server and its Windows Workflow Foundation component of .NET Framework 3.0. Oracle, IBM, Software AG, TIBCO Software, and others also provide business rules features for their BPM platforms. The business rules features that BPM platforms provide, however, tend to offer fewer tools and features than the major business rules platforms.
- **ETL platforms.** Vendors of extract, transform, and load (ETL) products for data warehousing also incorporate business rules. Examples include IBM, Informatica, SAP Business Objects, and Oracle. These products do not tend to employ the business rules products that we analyzed in this Forrester Wave™.

Customers Have Many Choices Of Business Rules Platforms

The platform market has several well-established vendors and a host of very strong challengers.

- **Fair Isaac and ILOG are the best-known vendors.** When clients approach Forrester for advice about business rules platforms, most are already using either Fair Isaac Blaze Advisor or ILOG JRules or know of these products. The high customer awareness that Fair Isaac and ILOG enjoy is fueled by the long histories of Blaze Advisor and ILOG JRules, the strong support among systems integrators for these products, and the marketing activities of these two vendors.
- **Corticon, Haley, InRule Technology, and Pegasystems often challenge Fair Isaac and ILOG.** These four vendors have strong products (Haley Limited has two) and often successfully compete with Fair Isaac and ILOG for business. The Corticon, Haley, and Pegasystems products are fundamentally different from Fair Isaac's Blaze Advisor and ILOG JRules. Corticon created a groundbreaking environment for business analysts and now has expanded into developer tools and SOA. Haley provides the most advanced natural language authoring facilities in the market,

along with both a general-purpose platform and a platform tuned for automating entitlement applications.⁵ InRule provides a comprehensive platform for shops that prefer to work with Microsoft's platforms. And whereas Blaze Advisor and ILOG JRules built their positions by appealing to programmers, PegaRULES appeals to application developers and seeks to hide the details of systems programming behind visual tools.

- **CA and SAP are in transition.** CA's Aion Business Rules Expert (BRE) is one of the oldest names in business rules platforms, but it has "fallen off the radar" for many clients. This could be the result of CA's shift in branding and in strategy over the past several years. CA now is concentrating on Aion BRE's strong position on the mainframe and is enjoying the benefits of the IBM zSeries' apparent resurgence. YASU QuickRules from SAP, acquired along with YASU Technologies during late 2007, will soon become part of the NetWeaver 7.1 platform suite as SAP NetWeaver Business Rules Management (SAP NetWeaver BRM).⁶ It will be available to customers as a licensed add-on with the SAP NetWeaver Composition Environment.
- **Experian, Innovations Software Technology, and Intelligent Results offer good alternatives.** And then there are the wildcards, each of which is new to Forrester's coverage of business rules platforms. Experian and Intelligent Results are large financial services applications specialists that also sell business rules platforms. Experian's Strategy Management Generation 3 (SMG3) and Intelligent Results Predigy are designed to allow professionals in risk management, collections, marketing, and other specialties to create and manage their own applications. Innovations Software Technology provides a general-purpose platform for Java shops that features powerful authoring tools and smooth integration with J2EE application servers.

Business rules platforms for custom application development have made only small inroads into the other major product categories. Some of the business rules platforms vendors also sell specialized applications and/or BPM (see Figure 1). In addition:

- **ILOG and Haley Limited have strong ISV followings.** ILOG has the largest following of independent software vendors (ISVs), companies that embed JRules in their solutions. Haley Limited has built deep ISV relationships with three of the four largest enterprise software vendors, Microsoft, Oracle, and SAP.
- **None of the business rules platform vendors tackles ETL.** However, some offer predictive analysis and other "business analytics" features.

For customers, this means that the skills they build on business rules platforms aren't necessarily transferable to other product categories that also use the technology. Industry consolidation is likely to change this fact as vendors selling suites of products begin to employ common business rules platforms across their products to reduce costs and simplify usage.

Figure 1 The Scope Of The Business Rules Platform Vendor Strategies

	Business rules	BPM	Business analytics
CA	✓	✓	
Corticon Technologies	✓		
Experian	✓	✓	✓
Fair Isaac	✓	✓	✓
Haley Limited	✓		
ILOG	✓		
Innovations Software Technology	✓	✓	
InRule Technology	✓		
Intelligent Results	✓	✓	✓
Pegasystems	✓	✓	✓
SAP	✓	✓	✓

39088

Source: Forrester Research, Inc.

Selecting A Rules Algorithm Means Making Tradeoffs

A common refrain in the business rules market is “Rete doesn’t matter.”⁷ Some vendors say that Rete inferencing algorithms are essential to be a true business rules platform. Other vendors say that you can solve any problem using a sequential algorithm. Both views are too extreme. Most business problems can be solved using either inferencing or sequential algorithms, but each has tradeoffs in rule design and in execution performance.

- **Sequential algorithms.** Sequential algorithms work like flowcharts. Authors explicitly connect rules to one another within their definitions. Thus, developers can trace the execution path through a group of rules. Sequential algorithms allow rules to be executed more than once, but only if a developer specifies this behavior using a looping construct. The advantage of sequential algorithms: Execution speed is fast, and developers explicitly determine the sequence of rules resulting in a transparent design. The disadvantage: Developers do not have the flexibility of designing rules independently of one another, which can raise design complexity.

- **Inferencing algorithms.** Rules processed by an inferencing algorithm are defined independently of one another. Unlike sequential rules, where the relationship between rules is defined or calculated explicitly, inferencing determines the dependencies between rules using data. The data drives the path through the rules rather than an explicit sequence. An inferencing algorithm may process rules more than once if a rule's execution changes data that affects the condition of another rule. The advantage of inferencing algorithms: Developers have the flexibility to design rules that are independent of one another, and rule interaction is driven by session data during execution. The disadvantage: For large rule sets, performance can be slow, and rules can interact with one another in unexpected ways.
- **Extended sequential algorithms.** Extended sequential algorithms seek to bridge the gap between sequential and inferencing processing. Like sequential, an extended sequential algorithm follows a predefined path through the rules. Unlike sequential, extended sequential algorithms can create that path by constructing a dependency graph. For example, if two rule authors separately define rules that apply to the same data, an extended sequential algorithm can examine both sets of rules and merge them into one sequential rule set. The advantage of extended sequential algorithms: Developers can obtain some of the flexibility of inferencing algorithms like Rete with lower performance risks. The disadvantage: Although rules can be defined independently like in inferencing, the rule execution path is sequential tempering the flexibility of the rules.

Six of the 13 platforms that we evaluated provide both inferencing and sequential algorithms, so developers get to choose which algorithm to use. The remaining seven platforms provide sequential or extended sequential algorithms but no inferencing algorithms.

BUSINESS RULES PLATFORMS EVALUATION OVERVIEW

Forrester included 11 vendors in the assessment: CA, Corticon Technologies, Experian, Fair Isaac, Haley Limited, ILOG, Innovations Software Technology, InRule Technology, Intelligent Results (which was recently acquired by First Data), Pegasystems, and SAP. Haley and ILOG each offer two platforms: Haley Expert Rules and Haley Office Rules/Determinations Engine, and ILOG JRules and Rules for .NET, respectively. Each of these vendors offers business rules platforms that (see Figure 2):

- **Employ business rules as their primary method for custom application development.** Forrester excluded from the evaluation BPM products that provide rules engines. These products do not tend to provide a full range of rules authoring and management features.
- **Support the full application life-cycle.** The business rules platforms that we evaluated incorporate application development tools, runtime environments, and operational management facilities. Most, but not all, of the products we evaluated allow developers to use an inferencing algorithm (such as Rete) at runtime.

- **Provide rules management facilities.** Each of the evaluated platforms incorporates change management features to help ensure that as business rules evolve, organizations maintain system integrity and correctness.

Forrester excluded open source business rules platforms, such as Red Hat JBoss Rules, primarily because they lack the authoring tools and platform management features typical of the platforms that we evaluated. Thus, the comparison would be unfair.⁸

Figure 2 Evaluated Vendors: Product Information And Selection Criteria

Vendor	Product evaluated	Product version evaluated	Version release date
CA	CA Aion Business Rules Expert	10.2	June 2006
Corticon Technologies	Corticon Business Rules Management System	5	October 2007
Experian	Strategy Management Generation 3	1.3 SP2	October 2007
Fair Isaac	Blaze Advisor Business Rules Management System	6.5	April 2007
Haley Limited	Haley Expert Rules with Haley Business Rules Engine and Haley Office Rules with Haley Determinations Engine	6.0	July 2007
		9.1	December 2007
ILOG	ILOG JRules and ILOG Rules for .NET	6.6	July 2007
		2.6	March 2007
Innovations Software Technology	Visual Rules	3.5.1	January 2007
InRule Technology	InRule	2.9.1	November 2007
Intelligent Results	Predigy	5	August 2007
Pegasystems	PegaRULES	5.3	June 2007
SAP	Yasu QuickRules from SAP	4.5 SP2	November 2007

Vendor selection criteria

Employ business rules as the primary method for custom application development

Support the full application life cycle

Provide rules management facilities

Source: Forrester Research, Inc.

After examining past and related research, assessing user needs and client inquiries, and completing vendor and expert interviews, we developed a comprehensive set of evaluation criteria for the business rules platform category (see Figure 3). We evaluated vendors and their products against approximately 175 criteria. We grouped the business rule platform criteria into three categories:

- **Current offering.** These criteria represent the features provided by the vendor within its business rules platform or platforms. The criteria address architectural characteristics; scalability; application development tools; tools for business analysts; life-cycle management; deployment, administration, and management; and interoperability. Current offering scores are the vertical axis on Forrester Wave diagrams. The Forrester Wave's current offering scores do not measure either product performance or stability and are not a proxy for benchmark tests for those purposes.
- **Strategy.** These criteria assess the vendor's product road map and other key product strategy elements, including the vendor's overall strategy, partner strategy, financial performance, pricing, and costs. Forrester's assessments of product and corporate strategy are subjective; the other scores are based on data. Strategy scores are the horizontal axis on the Forrester Wave diagrams.

Forrester's approach to the cost criteria in this section reflects the tension between vendor pricing policies and customer requirements in this market. Most vendors employ secret pricing schemes for their products, but customers want clear and predictable product costs. Thus, Forrester gave the highest scores to vendors that published their prices and low scores to vendors that did not. Forrester also rewarded vendors that provided a low-cost entry point for customers, which is another priority for our clients.

- **Market presence.** These criteria include the size of the vendor's employee and customer base, as well as its sales force and other channels. Market presence scores determine the size of the white circle surrounding each vendor's dot on the Forrester Wave diagram. The Forrester Wave's market presence scores do not measure market share and should not be used as a proxy for such estimates.

We used our evaluation criteria to assess the vendors and their products using the five sets of customer requirements that we observe in our work with customers. We've provided individual Forrester Wave diagrams for each of these points of view:

1. General-purpose platforms evaluated using a broad set of our criteria.
2. Specialized business rules platforms.
3. Business rules platforms for Java developers.

4. Business rules platforms for .NET developers.

5. Business rules platforms for business analysts.

Figure 3 Evaluation Criteria

CURRENT OFFERING

Platform architecture	How well is the platform architected?
Scalability features	Does the platform offer strong scalability features?
Development	What is the development model for the platform, and how well do the vendor's languages/tools support it?
Business analyst rules authoring tools	What tools for business analysts does the platform provide?
Business rules life-cycle management	What life-cycle management features does the platform provide for rules development?
Deployment, administration, and management	What tools and facilities does the platform provide to manage deployment, administration, and management of applications?
Standards and interoperability	What standards does the rules platform support? What third-party products can it interoperate with?

STRATEGY

Product strategy	How good is the vendor's product strategy for the platform?
Corporate strategy	How good is the vendor's corporate strategy?
Financial strength	How strong is the vendor financially?
Partner strategy effectiveness	How effective has the vendor's partner strategy been in creating a robust ecosystem for the product?
Cost	What is the cost of this product?

MARKET PRESENCE

Installed base	How large is the vendor's installed base of customers for this product and for all products?
Systems integrators	How many integrator partners have completed three or more deployments of any version of this product in the past 18 months?
Services	How strong are the vendor's implementation and training services?
Employees	How many engineers does the vendor have dedicated to this product? How big is the vendor's sales presence?

Source: Forrester Research, Inc.

The evaluations uncovered a market in which:

- **Corticon Technologies and ILOG are strong across the analyses.** Each of the vendors has extensive feature sets and strong strategies. ILOG has its roots in the artificial intelligence (AI) and expert systems movement of the 1980s, where the focus was on using inferencing algorithms to automate complex decisions and to make inferences from incomplete data. ILOG rules are defined declaratively and are often processed using the Rete inferencing algorithm. ILOG offers either an Eclipse-based integrated development environment (IDE) or a Visual Studio IDE for developers and a Web-based interface for business analysts.

Corticon Technologies is a Leader in three of the four Forrester Waves that measured general-purpose platforms. Release of the company's .NET offering later in 2008 will likely increase its stature in that segment of the business rules market.

- **Fair Isaac, Haley, and Pegasystems have strong features and fared well in some evaluations.** These three vendors each have feature sets that are as strong as or stronger than those of Corticon Technologies and ILOG.

Pegasystems' PegaRULES is a foundation for the company's BPM platform. Pegasystems rules are processed using an extended sequential algorithm. Pegasystems' interface is completely Web-based and incorporates many tools appropriate for business analysts.

Haley Expert and Pegasystems scored highest in the evaluation of tools for business analysts, and Haley Office Rules/Determinations Engine was also a strong option in that analysis.

Fair Isaac is a Leader in the Forrester Waves for platforms with the broadest feature set and for business analysts.

- **Innovations Software Technology and InRule Technology are on the move.** Innovations Software Technology is based in Germany and opened an office in Chicago in 2007. It is a Leader in platforms for Java developers and in the analysis of broad feature sets, and it scored well in the analysis of platforms for business analysts. Innovations' Visual Rules authoring environment uses a visual flowcharting metaphor to author rules.

InRule Technology has the strongest features of the pure .NET platforms, although the company's integration with Microsoft's Visual Studio is not yet complete. InRule's strategy scores were boosted by an extremely high growth rate.

- **SAP and CA are Strong Performers, but strategy is lacking.** SAP scored very well in the evaluation of platforms for Java developers but was held back by a lower strategy score that reflects the company's transition of YASU QuickRules to the NetWeaver platform.

CA's overall scores also reflect a product that has been in transition for the past two years. CA's renewed focus on Aion BRE strictly for decisioning applications and improved integration between the two parts of the product should drive improvement in future scores.

- **Haley is a leading specialized platform, but Experian and Intelligent Results also score high.** Haley Limited's Haley Office Rules has a unique rule authoring environment that cleverly uses Microsoft Word to extract rules from written documents. When coupled with the Haley Determinations Engine, the result is a strong platform for eligibility and other determination applications rooted in legislation and written regulations.⁹

Experian and Intelligent Results are Strong Performers in specialized platforms for financial decisioning. Experian and Intelligent Results shone as specialized platforms for financial decisioning applications such as credit decisioning and collections. Neither of these platforms is designed to be a general-purpose platform.

View No. 1: The Broadest Feature Sets

This analysis answers the question: Which business rules platforms provide the broadest sets of features? The analysis excludes three specialized platforms (Experian SMG3, Intelligent Results Predigy, and Haley Office Rules used with the Haley Determinations Engine), as these products are not designed to satisfy the broadest range of requirements. This analysis also excludes Java- and .NET-specific features. The evaluation uncovered a market in which (see Figure 4):

- **Pegasystems, Haley, Corticon, Fair Isaac, and ILOG, lead; Innovations is a borderline Leader.** Pegasystems' PegaRULES, Haley's Expert Rules, Corticon's Business Rules Management System, Fair Isaac Blaze Advisor, and ILOG's JRules offer broad and strong developer and business analyst tools; rules life-cycle management; and deployment, administration, and management features. They also earned their Leader positions due to strong scores for product architecture and strategy.

Haley Expert Rules — the new name for the product formerly known as Haley Authority/HaleyRules — is well known for its leading tools for business analysts, but it also earned high scores across the board in our feature categories.

Corticon Technologies' standing in this analysis reflects the broad expansion of the Corticon business rules platform for developers during the past 18 months. Corticon now provides a strong general-purpose platform and earned strong strategy scores, as well.

Innovations Software Technology, a newcomer to our research on this category, made a good showing on the strength of its features for developers and business analysts, as well as its solid Java-based architecture.

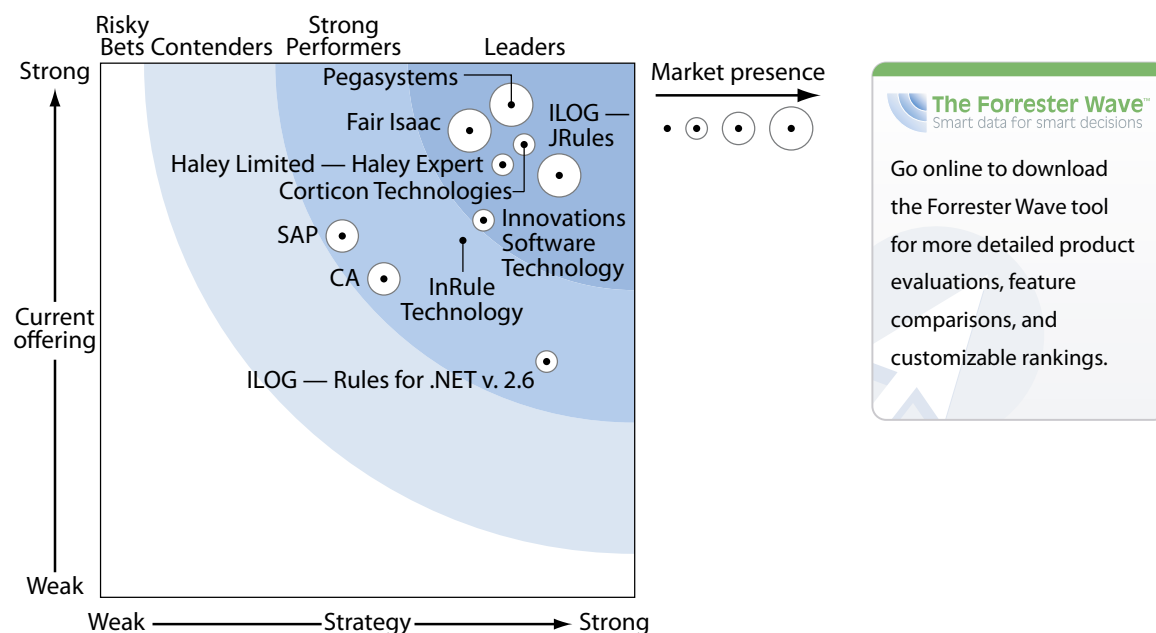
- **InRule is a Strong Performer, offering a very competitive option.** InRule Technology was within striking distance of the Leaders sector in this analysis. The InRule platform's scores reflect

the aggressive build-out of this company and its product during the past two years. InRule's progress is remarkable. InRule scored moderately well across the board in our feature criteria, indicating that the platform doesn't yet have the depth of features offered by the Leaders. InRule earned moderate strategy scores, which is typical of small companies, but was helped by its transparent pricing and its 100% growth rate.

- **CA and SAP are Strong Performers but lag behind the pack.** CA and SAP are both in transition with their business rules platforms. Still, both are Strong Performers in this analysis. Of the two, SAP earned the higher score, primarily due to stronger tools for developers and business analysts, as well as for scalability features and architecture. CA scored higher than SAP in rule life-cycle management and in standards and interoperability. CA earned higher strategy scores than SAP but lower scores for market presence (SAP's market presence is higher than CA's, although both are large).

Customers must approach both vendors and their business rules platforms with caution. CA is focusing more and more on its strengths in mainframe-based rules applications and is using the Web-based Rules Manager module of Aion as an adjunct to that position. If mainframe rules are not your priority, CA may not be an ideal choice. Meanwhile, SAP plans to offer NetWeaver Business Rules Management as an add-on component integrated into the NetWeaver Composition Environment. SAP customers will get a powerful new set of integrated business rules features, but customers who use technology platforms other than NetWeaver will be disappointed.

Figure 4 Forrester Wave™: Business Rules Platforms, Broadest Feature Sets, Q1 '08



Source: Forrester Research, Inc.

Figure 4 Forrester Wave™: Business Rules Platforms, Broadest Feature Sets, Q1 '08 (Cont.)

	Forrester's Weighting	CA	Corticon Technologies	Fair Isaac	Haley Limited — Haley Expert	ILOG — JRules	ILOG — Rules for .NET v. 2.6	Innovations Software Technology	InRule Technology	Pegasystems	SAP
CURRENT OFFERING	50%	2.97	4.22	4.34	4.03	3.92	2.18	3.51	3.32	4.59	3.36
Platform architecture	15%	3.35	4.42	5.00	4.30	4.40	3.43	3.55	3.38	4.03	3.84
Scalability features	10%	2.50	4.50	5.00	5.00	3.75	3.25	2.75	4.00	5.00	5.00
Development	20%	3.22	4.41	4.52	3.64	3.87	2.92	4.43	3.06	4.87	3.48
Business analyst rules authoring tools	20%	2.70	4.20	3.70	4.22	3.20	0.92	3.56	3.00	4.40	3.20
Business rules life-cycle management	15%	3.58	4.94	5.00	4.08	5.00	2.48	4.29	3.93	4.85	3.24
Deployment, administration, and management	10%	2.86	2.85	4.07	3.92	4.42	1.16	3.15	3.15	4.52	2.92
Standards and interoperability	10%	2.15	3.55	2.85	3.15	2.80	0.90	1.50	3.00	4.50	1.75
STRATEGY	50%	2.65	3.97	3.47	3.78	4.30	4.18	3.59	3.39	3.85	2.27
Product strategy	15%	1.70	4.30	3.00	4.10	4.10	4.10	3.30	4.30	4.50	1.10
Corporate strategy	15%	3.00	5.00	4.50	4.50	5.00	5.00	3.00	5.00	5.00	4.00
Financial strength	25%	3.20	2.50	3.50	3.60	4.40	4.40	3.00	2.20	3.80	2.90
Partner strategy effectiveness	15%	0.40	3.80	3.00	3.80	5.00	3.40	3.80	1.60	2.20	0.60
Cost	30%	3.60	4.60	3.40	3.40	3.60	4.00	4.40	4.00	3.80	2.30
MARKET PRESENCE	0%	3.26	2.30	4.44	2.91	4.41	2.95	2.90	1.95	4.40	3.84
Installed base	40%	3.40	2.20	4.10	1.90	3.95	2.80	2.45	3.05	3.70	2.10
Systems integrators	20%	3.00	3.00	5.00	3.00	5.00	0.00	3.00	0.00	5.00	5.00
Services	20%	1.50	3.00	4.00	4.00	5.00	5.00	4.00	3.00	5.00	5.00
Employees	20%	5.00	1.10	5.00	3.75	4.15	4.15	2.60	0.65	4.60	5.00

All scores are based on a scale of 0 (weak) to 5 (strong).

Source: Forrester Research, Inc.

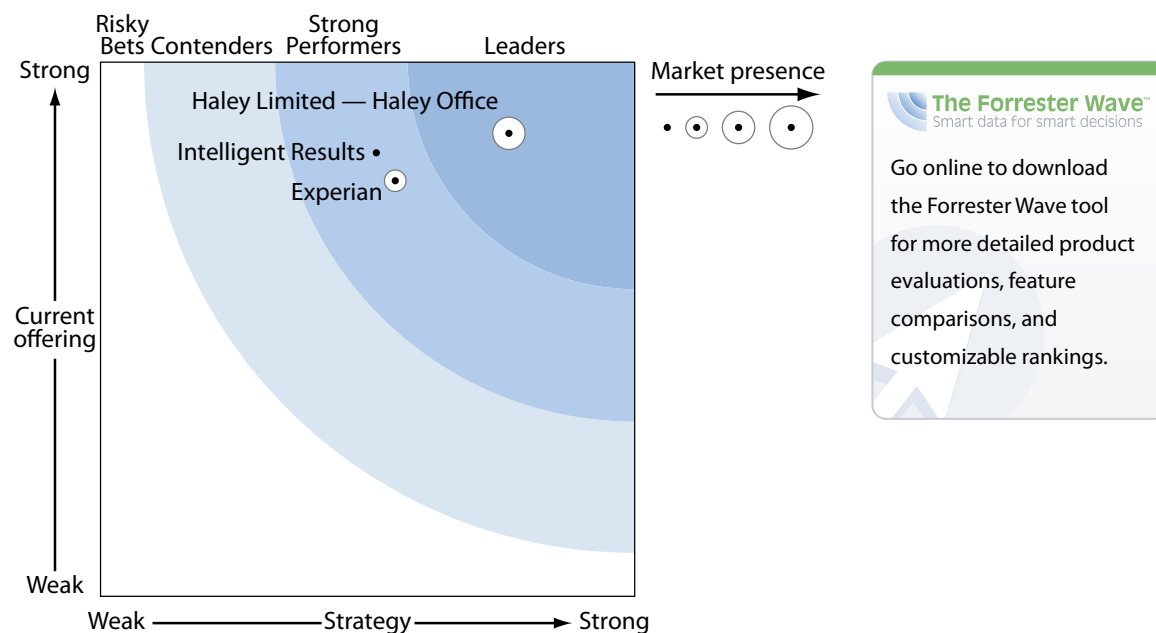
This evaluation of the business rules platform market is intended to be a starting point only. Readers are encouraged to view detailed product evaluations and adapt the criteria weightings to fit their individual needs through the Forrester Wave Excel-based vendor comparison tool.

View No. 2: Specialized Platform Vendors

Experian's SMG3, Intelligent Results Predigy, and Haley Office Rules, when used with the Haley Determinations Engine, shine for certain applications but are not intended to be used as a replacement for general-purpose developer platforms like Pegasystems' PegaRULES and ILOG's JRules. Experian and Intelligent Results (First Data) concentrate on decision-making in financial services, while Haley Office Rules is designed for eligibility applications. It is common for a large financial company to have one of these specialized platforms and also a general-purpose rules platform. From our evaluation, we found that (see Figure 5):

- **Haley Office Rules is a Leader (for eligibility determination applications).** Haley Office Rules is the only platform to offer rule authoring in Microsoft Word, which permits business analysts to extract rules directly from plain text legislation or policy documents. The platform also offers decision table authoring in Excel and rule-flow authoring in Visio. Haley Office Rules can automatically generate a Web-based interview page flow from defined rules. Haley's planned enhancements include support for decision-tree authoring in Visio and vertical solution modules for benefits eligibility determination and fraud detection.

Figure 5 Forrester Wave™: Specialized Business Rules Platforms, Q1 '08



Source: Forrester Research, Inc.

Figure 5 Forrester Wave™: Specialized Business Rules Platforms, Q1 '08 (Cont.)

	Forrester's Weighting	Experian	— Haley Limited Haley Office	Intelligent Results
CURRENT OFFERING	50%	3.87	4.30	4.13
Platform architecture	10%	3.33	3.03	3.35
Scalability features	20%	4.60	5.00	5.00
Development	10%	3.35	3.78	3.82
Business analyst rules authoring tools	30%	4.20	4.50	4.60
Business rules life-cycle management	10%	4.65	4.45	4.15
Deployment, administration, and management	10%	4.55	3.25	3.05
Standards and interoperability	10%	1.00	5.00	3.10
STRATEGY	50%	2.77	3.83	2.58
Product strategy	15%	2.90	4.50	2.40
Corporate strategy	15%	3.00	4.50	3.50
Financial strength	40%	4.40	3.60	3.80
Partner strategy effectiveness	5%	0.00	3.80	0.00
Cost	25%	0.50	3.40	0.70
MARKET PRESENCE	0%	2.76	3.39	1.64
Installed base	40%	3.40	3.10	1.85
Systems integrators	20%	0.00	3.00	0.00
Services	20%	4.00	4.00	2.00
Employees	20%	3.00	3.75	2.50

All scores are based on a scale of 0 (weak) to 5 (strong).

Source: Forrester Research, Inc.

• **Experian's SMG3 and Intelligent Results' Predigy are strong in financial risk management.**

Experian and Intelligent Results provide business rules platforms that are tailored for risk managers to provide financial risk management and credit decisioning applications. Both Experian and Intelligent Results are known for offering a broad range of financial and credit services to financial institutions and credit card companies, and they use the platforms within their own company. Experian redesigned its SMG3 product a few years ago to run on the Java platform that includes a Java Swing rule authoring and management IDE. The SMG3 authoring

environment provides many authoring metaphors that would be immediately familiar to risk managers such as Champion/Challenger.

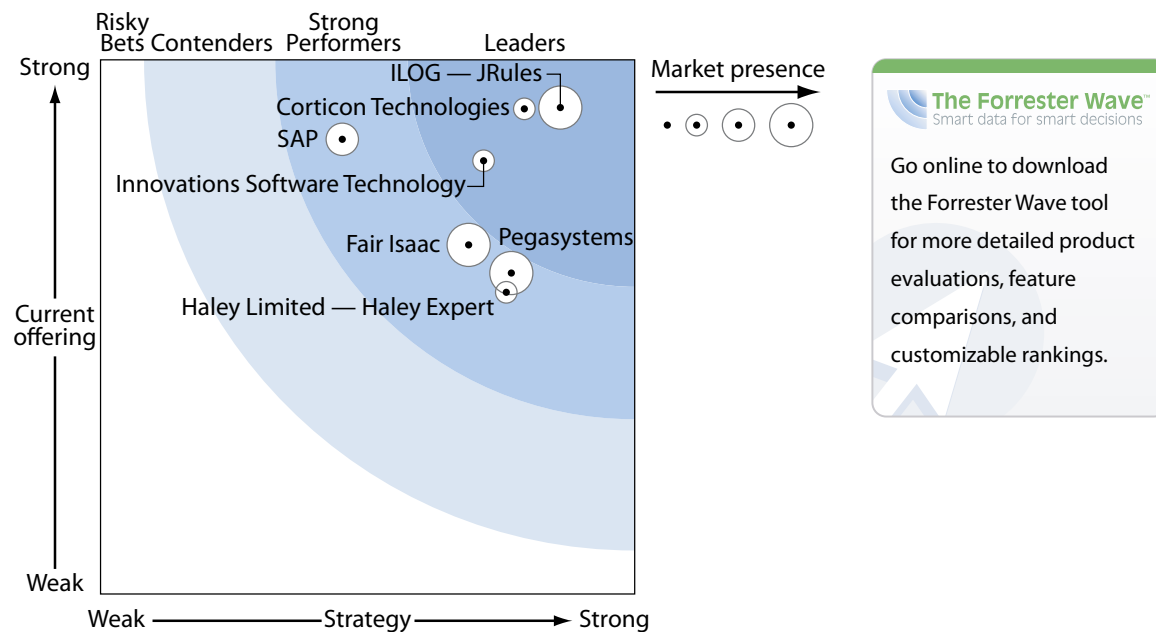
Predigy customers use a browser-based IDE to author and manage rules and data sets. Risk managers and marketing specialists will be especially comfortable with the built-in financial modeling capabilities. Intelligent Results also offers a hosted version of Predigy that is integrated with First Data's credit information services.

This evaluation of the business rules platform market is intended to be a starting point only. Readers are encouraged to view detailed product evaluations and adapt the criteria weightings to fit their individual needs through the Forrester Wave Excel-based vendor comparison tool.

View No. 3: Business Rules Platforms For Java Developers

This analysis zeroes in on the products that Forrester would recommend to application development managers seeking platforms for Java/J2EE/Java EE developers. The analysis places heavy emphasis on each platform's support of either the Eclipse or the NetBeans IDEs, as well as its support for J2EE platform standards, that such shops value. For Java developers (see Figure 6):

Figure 6 Forrester Wave™: Business Rules Platforms For Java Developers, Q1 '08



Source: Forrester Research, Inc.

Figure 6 Forrester Wave™: Business Rules Platforms For Java Developers, Q1 '08 (Cont.)

	Forrester's Weighting	Corticon Technologies	Fair Isaac	Haley Limited — Haley Expert	ILOG — JRules	Innovations Software Technology	Pegasystems	SAP
CURRENT OFFERING	50%	4.54	3.24	2.79	4.53	4.04	2.98	4.24
Platform architecture	10%	4.37	5.00	3.79	4.18	4.06	4.65	3.67
Scalability features	5%	4.50	5.00	5.00	3.75	2.75	5.00	5.00
Development	45%	4.97	1.20	0.98	4.45	4.93	1.20	4.38
Business analyst rules authoring tools	10%	2.75	5.00	4.10	5.00	0.50	2.75	5.00
Business rules life-cycle management	10%	4.93	5.00	3.57	5.00	3.89	4.76	3.21
Deployment, administration, and management	10%	3.85	4.73	4.58	4.48	3.85	4.78	3.78
Standards and interoperability	10%	4.90	4.75	4.90	4.70	4.50	5.00	4.55
STRATEGY	50%	3.97	3.47	3.78	4.30	3.59	3.85	2.27
Product strategy	15%	4.30	3.00	4.10	4.10	3.30	4.50	1.10
Corporate strategy	15%	5.00	4.50	4.50	5.00	3.00	5.00	4.00
Financial strength	25%	2.50	3.50	3.60	4.40	3.00	3.80	2.90
Partner strategy effectiveness	15%	3.80	3.00	3.80	5.00	3.80	2.20	0.60
Cost	30%	4.60	3.40	3.40	3.60	4.40	3.80	2.30
MARKET PRESENCE	0%	2.30	4.44	2.91	4.41	2.90	4.40	3.84
Installed base	40%	2.20	4.10	1.90	3.95	2.45	3.70	2.10
Systems integrators	20%	3.00	5.00	3.00	5.00	3.00	5.00	5.00
Services	20%	3.00	4.00	4.00	5.00	4.00	5.00	5.00
Employees	20%	1.10	5.00	3.75	4.15	2.60	4.60	5.00

All scores are based on a scale of 0 (weak) to 5 (strong).

Source: Forrester Research, Inc.

- **ILOG, Corticon, and Innovations are Leaders.** ILOG, Corticon Technologies, and Innovations Software Technology all provide business rule authoring plug-ins for an Eclipse-based IDE. Thus, a Java developer can easily integrate a Java execution object model (XOM) and learn each product's rule-authoring tools within a familiar IDE. Each platform permits rule applications to be deployed to J2EE host containers.

- **SAP has strong features.** Like the Leaders in this evaluation, YASU QuickRules from SAP offers an Eclipse-based IDE for rules development and has a strong range of developer features. As noted above, SAP's strategy scores suffered from the transition of YASU QuickRules from SAP from a standalone platform to a component of NetWeaver.
- **Fair Isaac, Pegasystems, and Haley are Strong Performers.** Fair Isaac's Blaze Advisor, PegaRULES, and Haley Expert Rules are implemented in Java, but they lack support for an Eclipse-based or NetBeans IDE. PegaRULES offers a rich array of development tools in a custom Internet client. Haley Expert Rules provides developers with a custom Windows-based client IDE alongside its natural language authoring tool. Blaze Advisor offers a custom Windows-based client IDE, but Fair Isaac plans to provide an Eclipse-based IDE in a future release. All three products allow developers to import and export Java classes.

This evaluation of the business rules platform market is intended to be a starting point only. Readers are encouraged to view detailed product evaluations and adapt the criteria weightings to fit their individual needs through the Forrester Wave Excel-based vendor comparison tool.

View No. 4: Business Rules Platforms For .NET Developers

This analysis concentrates on the products that Forrester would recommend to application development managers seeking platforms for .NET developers. Two Forrester Waves show the choices available. The first identifies the two pure .NET vendors (ILOG Rules for .NET 2.6 and InRule) on the "broadest feature set" analysis. The second evaluation focuses on support for "pure .NET" development and deployment among platforms that can be used with Microsoft's platform (see Figure 7). The key questions in the second evaluation: Do that platform's tools run as Visual Studio plug-ins, and does the platform have support for .NET assemblies for deployment? Forrester chose to focus on these contemporary .NET platform formats because clients much prefer them to older formats, including ASP and COM+.

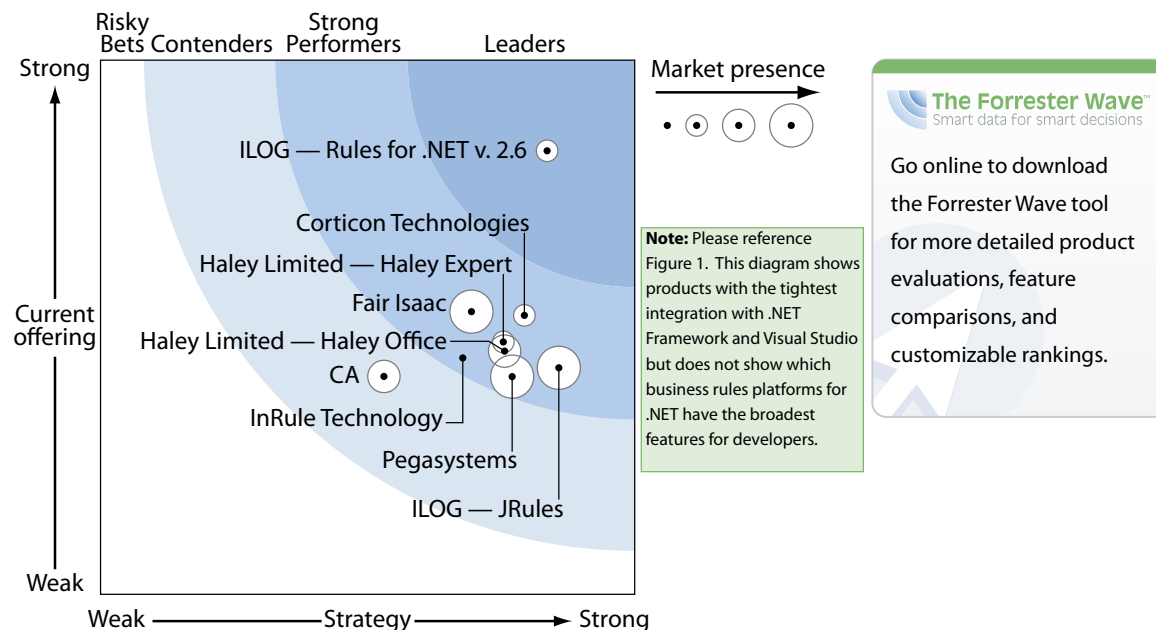
- **ILOG Rules for .NET 2.6 and InRule are the only platforms implemented in native .NET.** ILOG's Rules for .NET 2.6 is very well integrated with Visual Studio and the .NET Framework and common language runtime (CLR) but has relatively weak features. Rules for .NET 3.0, due out later in 2008, promises to significantly expand the features of ILOG's product for .NET, adding a JRules-class runtime engine and better testing tools.

InRule Technology has strong features — it is within striking distance of the Leaders sector — but lacks an IDE integrated with Visual Studio. InRule's product road map includes more integration with Microsoft SharePoint and Visual Studio, internationalization, and enhanced regression testing tools.

- **CA, Corticon, Fair Isaac, and Haley support deployment to .NET Framework.** None of these platforms provides an IDE that is integrated with Visual Studio. CA, Fair Isaac, and Haley support deployment to .NET as .NET assemblies. Corticon plans to offer customers a .NET deployment option during the first half of 2008.
- **ILOG provides a .NET native execution engine with JRules.** ILOG's JRules offers both a native Java and a native .NET rules execution engine, allowing the customer to deploy rules to either engine. However, JRules IDE is based on Java and cannot be integrated with Visual Studio.
- **Pegasystems does not support .NET Framework deployment directly.** Pegasystems provides customers with its own IDE and does not support integration with Visual Studio. Customers can deploy PegaRULES applications as COM+ objects, an older format that can be integrated with .NET. Pegasystems does plan to provide rule authoring capabilities in Microsoft Visio in a future release.

This evaluation of the business rules platform market is intended to be a starting point only. Readers are encouraged to view detailed product evaluations and adapt the criteria weightings to fit their individual needs through the Forrester Wave Excel-based vendor comparison tool.

Figure 7 Forrester Wave™: Business Rules Platforms For .NET Developers, Q1 '08



Source: Forrester Research, Inc.

Figure 7 Forrester Wave™: Business Rules Platforms For .NET Developers, Q1 '08 (Cont.)

	Forrester's Weighting	CA	Corticon Technologies	Fair Isaac	Haley Limited — Haley Expert	Haley Limited — Haley Office	ILOG — JRules	ILOG — Rules for .NET v. 2.6	InRule Technology	Pegasystems
CURRENT OFFERING	50%	2.02	2.59	2.62	2.35	2.27	2.10	4.13	2.20	2.01
Platform architecture	10%	3.15	3.78	5.00	4.40	1.68	2.60	4.18	3.95	2.63
Scalability features	0%	2.50	4.50	5.00	5.00	3.75	3.75	3.25	4.00	5.00
Development	50%	0.76	1.22	1.20	0.98	0.58	0.70	4.32	0.82	1.20
Business analyst rules authoring tools	10%	0.50	5.00	0.50	0.50	5.00	0.50	4.50	0.50	5.00
Business rules life-cycle management	10%	3.58	4.94	5.00	4.08	4.01	5.00	2.48	3.93	4.85
Deployment, administration, and management	10%	4.49	2.85	4.98	4.68	4.15	4.73	4.19	4.75	0.88
Standards and interoperability	10%	4.70	3.20	4.75	4.90	5.00	4.70	4.40	4.75	0.75
STRATEGY	50%	2.65	3.97	3.47	3.78	3.78	4.30	4.18	3.39	3.85
Product strategy	15%	1.70	4.30	3.00	4.10	4.10	4.10	4.10	4.30	4.50
Corporate strategy	15%	3.00	5.00	4.50	4.50	4.50	5.00	5.00	5.00	5.00
Financial strength	25%	3.20	2.50	3.50	3.60	3.60	4.40	4.40	2.20	3.80
Partner strategy effectiveness	15%	0.40	3.80	3.00	3.80	3.80	5.00	3.40	1.60	2.20
Cost	30%	3.60	4.60	3.40	3.40	3.40	3.60	4.00	4.00	3.80
MARKET PRESENCE	0%	3.26	2.30	4.44	2.91	3.39	4.41	2.95	1.95	4.40
Installed base	40%	3.40	2.20	4.10	1.90	3.10	3.95	2.80	3.05	3.70
Systems integrators	20%	3.00	3.00	5.00	3.00	3.00	5.00	0.00	0.00	5.00
Services	20%	1.50	3.00	4.00	4.00	4.00	5.00	5.00	3.00	5.00
Employees	20%	5.00	1.10	5.00	3.75	3.75	4.15	4.15	0.65	4.60

All scores are based on a scale of 0 (weak) to 5 (strong).

Source: Forrester Research, Inc.

View No. 5: Business Rules Platforms For Business Analysts

The last of our evaluations answers the question: Which business rules platforms have the strongest tools for business analysts? The analysis focuses on each platform's tools for business analysts, as well as its support for change management and testing so that business analyst work is coordinated with work by application developers (see Figure 8). Because Haley Office Rules/Determinations Engine is designed for business analysts, we included this product in our evaluation, even though it is a specialized platform. Our evaluation revealed that:

- **Pegasystems, Haley, Corticon, Fair Isaac, and ILOG's JRules lead.** These vendors score well because of the number of tools for business analysts that they offer. Pegasystems' PegaRULES provides a browser-based IDE that contains tools usable by both business analysts and developers. PegaRULES' tools designed for business analysts include decision tables, decision trees, decision maps, declarative expressions, triggers, graphical rule flows, and events. Pegasystems also provides 20 prebuilt domain models and rule sets for PegaRULES and its SmartBPM Suite.

Haley offers customers the choice of two different natural language authoring environments — Haley Expert Rules and Haley Office Rules.

Corticon has breakthrough tools for business analysts; its advanced decision table is proven in a range of decisioning applications.

Fair Isaac provides a broad range of tools for business analysts within its Web-based rules maintenance application, including full testing features, as well as within its IDE.

ILOG has made great progress in tools for business analysts during the past two years. JRules Rule Team Server provides a Web-based interface for rule authoring and maintenance by business analysts, and JRules Rule Scenario Manager is a Web-based tool providing full testing features for both business analysts and developers. ILOG provides embeddable authoring controls for application-specific decision table authoring.

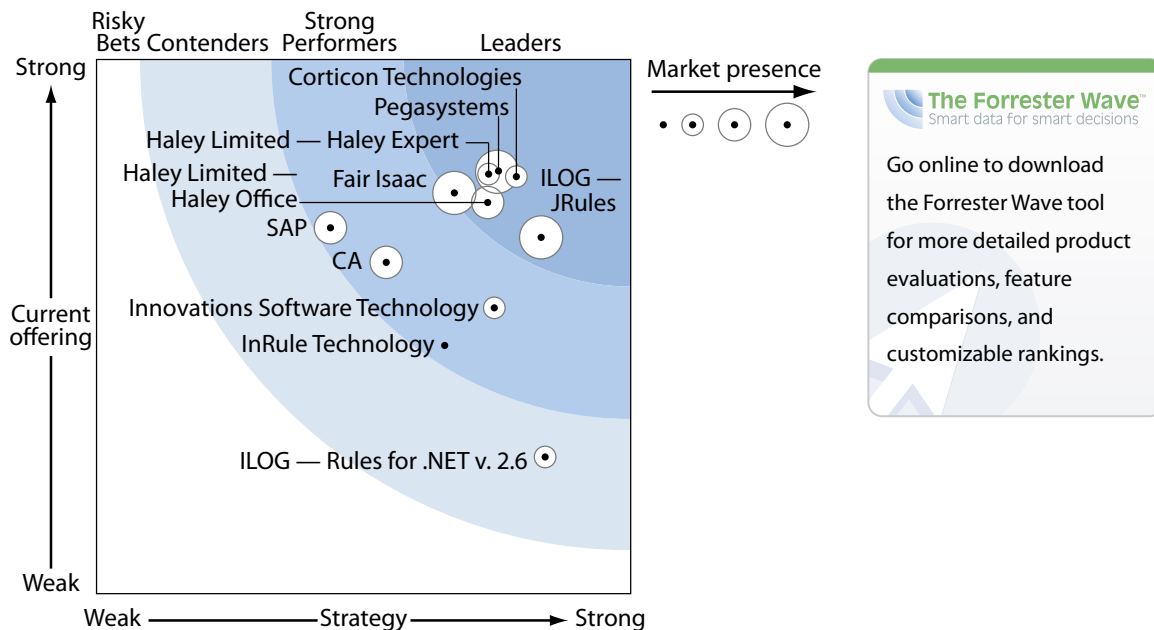
- **CA and SAP are Strong Performers.** CA and SAP both provide environments for business analysts as part of their platforms. YASU QuickRules' analytics tool from SAP is designed specifically for authoring and testing by business analysts and includes simulation features. YASU QuickRules from SAP also provides a Web-based rules maintenance application and authoring in Excel. The Rules Manager module of CA Aion BRE provides innovative Web-based authoring and testing facilities but needs stronger integration with the Aion BRE component of CA's platform.
- **Innovations and InRule are Strong Performers but offer fewer tools for business analysts.** Innovations Software Technology's Visual Rules enables rule authors to define sequential rules using a graphical tool that looks much like a flowchart. It is a powerful tool for application

developers but is not ideal for business analysts. Innovations plans to provide tools for business analysts during 2008. InRule Technology's environment is primarily aimed at developers, but it offers embeddable rule-authoring controls for both Windows and Web applications that allow business analysts to customize existing applications.

- **ILOG Rules for .NET is a Contender.** Although Rules for .NET allows for rules authoring in Word and Excel, business analysts would be forced to use Visual Studio to import and manage rule projects. ILOG's road map includes plans to expand tools for business analysts and to share the collaboration tools that are currently offered only exclusively in JRules Rule Team Server.

This evaluation of the business rules platform market is intended to be a starting point only. Readers are encouraged to view detailed product evaluations and adapt the criteria weightings to fit their individual needs through the Forrester Wave Excel-based vendor comparison tool.

Figure 8 Forrester Wave™: Business Rules Platforms For Business Analysts, Q1 '08



Source: Forrester Research, Inc.

Figure 8 Forrester Wave™: Business Rules Platforms For Business Analysts, Q1 '08 (Cont.)

	Forrester's Weighting	CA	Corticon Technologies	Fair Isaac	Haley Limited — Haley Expert	Haley Limited — Haley Office	ILOG — JRules	ILOG — Rules for .NET v. 2.6	Innovations Software Technology	InRule Technology	Pegasystems	SAP
CURRENT OFFERING	50%	3.08	3.86	3.74	3.90	3.65	3.31	1.25	2.65	2.30	3.93	3.40
Platform architecture	10%	3.56	4.24	5.00	3.64	2.99	4.92	4.23	2.92	2.73	4.10	4.08
Scalability features	0%	2.50	4.50	5.00	5.00	3.75	3.75	3.25	2.75	4.00	5.00	5.00
Development	5%	3.21	4.74	4.69	4.06	2.03	2.65	2.69	4.86	2.99	4.90	2.51
Business analyst rules authoring tools	55%	3.02	3.70	3.40	4.14	4.40	3.02	0.76	2.56	1.70	3.30	3.34
Business rules life-cycle management	15%	4.58	4.88	5.00	2.88	2.58	5.00	1.23	2.88	3.68	5.00	3.68
Deployment, administration, and management	5%	1.80	2.70	4.70	4.70	2.30	3.00	1.70	1.75	2.70	4.20	2.30
Standards and interoperability	10%	1.25	3.00	1.50	3.85	3.25	1.25	0.00	1.85	2.55	5.00	3.65
STRATEGY	50%	2.71	3.93	3.36	3.67	3.67	4.16	4.20	3.73	3.26	3.76	2.19
Product strategy	15%	1.70	4.30	3.00	4.10	4.10	4.10	4.10	3.30	4.30	4.50	1.10
Corporate strategy	5%	3.00	5.00	4.50	4.50	4.50	5.00	5.00	3.00	5.00	5.00	4.00
Financial strength	25%	3.20	2.50	3.50	3.60	3.60	4.40	4.40	3.00	2.20	3.80	2.90
Partner strategy effectiveness	15%	0.40	3.80	3.00	3.80	3.80	5.00	4.20	3.80	1.40	2.40	1.20
Cost	40%	3.60	4.60	3.40	3.40	3.40	3.60	4.00	4.40	4.00	3.80	2.30
MARKET PRESENCE	0%	3.26	2.30	4.44	2.55	3.39	4.41	2.95	2.90	1.95	4.40	3.84
Installed base	40%	3.40	2.20	4.10	1.00	3.10	3.95	2.80	2.45	3.05	3.70	2.10
Systems integrators	20%	3.00	3.00	5.00	3.00	3.00	5.00	0.00	3.00	0.00	5.00	5.00
Services	20%	1.50	3.00	4.00	4.00	4.00	5.00	5.00	4.00	3.00	5.00	5.00
Employees	20%	5.00	1.10	5.00	3.75	3.75	4.15	4.15	2.60	0.65	4.60	5.00

All scores are based on a scale of 0 (weak) to 5 (strong).

Source: Forrester Research, Inc.

VENDOR PROFILES

Leaders

- **ILOG.** With an installed based of 984 customers and 4,542 licenses sold, ILOG has the most market penetration of any of the vendors evaluated. ILOG offers both JRules and Rules for .NET and has a strong and comprehensive road map. Planned enhancements for JRules include expanded tools for business analysts, including the addition of testing to JRules Rule Team Server and authoring using Microsoft Word and Excel, and a new Agile business rules development methodology. Planned enhancements for Rules for .NET include free-text editing of business rules and decision tables in Microsoft Office 2007, as well as the addition of a Rule Execution Server for managed execution and rules services.
- **Fair Isaac.** Fair Isaac's Blaze Advisor is used by 406 customers in more than 500 installations. Fair Isaac has a strong and comprehensive road map for Blaze Advisor. Planned enhancements include improved usability for the rules maintenance application using rich Internet application (RIA) technology, improved deployment management, and a new Eclipse-based IDE. Fair Isaac offers ASP services that customers can use to host their Blaze Advisor applications.
- **Pegasystems.** Pegasystems has more than 200 customers with more than 300 production applications using all versions of PegaRULES. Fifty-four customers have bought or upgraded PegaRULES in 2007. Pegasystems has a strong and comprehensive road map for PegaRULES. Planned enhancements include easing the direct capture of business objectives in metadata, a new automated testing framework, and multivariant and multicircumstance decisioning.
- **Corticon Technologies.** Corticon has an installed base of 250 customers and reported revenue growth in excess of 50% during the past four years. Corticon has a strong and comprehensive road map. Planned enhancements to Corticon include the addition of predictive analysis, event correlation, and rules optimization features, as well as integration of Corticon's modeling studio with selected third-party rules engines.
- **Haley Limited.** Haley Limited was formed in 2007 when RuleBurst acquired Haley Systems. Offering two different but complementary products, Haley has a strong and comprehensive road map for both Haley Office Rules and Haley Expert Rules. Haley plans to provide a hosted service in the future. Haley did not disclose its installed based of customers for either of its platforms but has disclosed its work with the Internal Revenue Service (IRS) in the US and HM Revenue & Customs in the UK, as well as many governmental agencies in Australia, the location of its headquarters.
- **Innovations Software Technology.** Innovations Software Technology is one of the newcomers to our research on this market. With a strong feature set and very attractive pricing, Innovations has managed to add 70 of its total 100 customers during the past 12 months, most of which are

based in Europe and international markets. Innovations has a strong and comprehensive road map for Visual Rules, including a Web-based rules browser, reporting server, and workflow.

Strong Performers

- **InRule Technology.** InRule claimed revenue of \$4 million during its past four quarters — a 100% growth rate. InRule's success is due to two factors: its strong product development and its focus on the midmarket with a product based on Microsoft's stack. InRule has a strong and comprehensive road map for its platform. InRule continues to grow and has an installed base of 101 customers.
- **Experian.** Experian has an installed base of 395 customers, 179 of which are on the latest version of SMG3. Customers using SMG3 include banks, financial services, and telecom companies in more than 60 countries across the globe and include the majority of the leading financial institutions in each market. Experian's market presence, strategy, and planned enhancements for predictive analytics will ensure that it is a strong choice in the business rules market for years to come.
- **Intelligent Results.** First Data's recent acquisition of Intelligent Results and strong strategy should help Predigy grow beyond the current 18 customers that are now using the platform. Predigy's browser-based interface and hosted offering should help Intelligent Results be more competitive. First Data has approximately nine dedicated sales resources focused on the US. The company is in the early stages of having the larger First Data global sales teams carry Predigy into their accounts.

Contenders

- **CA.** CA now markets Aion BRE primarily for mainframe-based applications, although Aion BRE is also available for other platforms. CA's road map includes improved integration of the two components of Aion BRE — the IDE and Rules Manager. CA has 750 customers for all versions of Aion BRE. Several large customers use Aion BRE, including State Farm Mutual Automobile Insurance.
- **SAP.** SAP's road map includes plans to integrate YASU QuickRules from SAP into a new NetWeaver Business Rules Management add-on for the NetWeaver Composition environment starting with NetWeaver version 7.1. Customers will be able to deploy rule services within the NetWeaver CE runtime that can be consumed by ABAP business applications, BPM, and other processes. SAP also plans to enhance the product with additional rules authoring tools, rules management, and real-time event processing capabilities. In the long term, SAP has the opportunity to use YASU QuickRules from SAP as its single business rules engine within its enterprise applications, replacing the several rules and calculation engines that it employs today.

SUPPLEMENTAL MATERIAL

Online Resource

The online version of Figures 4, 5, 6, 7, and 8 are Excel-based vendor comparison tools that provide detailed product evaluations and customizable rankings.

Data Sources Used In This Forrester Wave

Forrester used a combination of four data sources to assess the strengths and weaknesses of each solution:

- **Hands-on lab evaluations.** Vendors spent one day with a team of analysts who performed a hands-on evaluation of the product using a scenario-based testing methodology. We evaluated each product using the same scenario(s), creating a level playing field by evaluating every product on the same criteria.
- **Vendor surveys.** Forrester surveyed vendors on their capabilities as they relate to the evaluation criteria. Once we analyzed the completed vendor surveys, we conducted vendor calls where necessary to gather details of vendor qualifications.
- **Product demos.** We asked vendors to conduct demonstrations of their product's functionality. We used findings from these product demos to validate details of each vendor's product capabilities.
- **Customer reference calls.** To validate product and vendor qualifications, Forrester also conducted reference calls with two of each vendor's current customers.

The Forrester Wave Methodology

We conduct primary research to develop a list of vendors that meet our criteria to be evaluated in this market. From that initial pool of vendors, we then narrow our final list. We choose these vendors based on: 1) product fit; 2) customer success; and 3) Forrester client demand. We eliminate vendors that have limited customer references and products that don't fit the scope of our evaluation.

After examining past research, user need assessments, and vendor and expert interviews, we develop the initial evaluation criteria. To evaluate the vendors and their products against our set of criteria, we gather details of product qualifications through a combination of lab evaluations, questionnaires, demos, and/or discussions with client references. We send evaluations to the vendors for their review, and we adjust the evaluations to provide the most accurate view of vendor offerings and strategies.

We set default weightings to reflect our analysis of the needs of large user companies — and/or other scenarios as outlined in the Forrester Wave document — and then score the vendors based on a clearly defined scale. These default weightings are intended only as a starting point, and readers are

encouraged to adapt the weightings to fit their individual needs through the Excel-based tool. The final scores generate the graphical depiction of the market based on current offering, strategy, and market presence. Forrester intends to update vendor evaluations regularly as product capabilities and vendor strategies evolve.

ENDNOTES

- ¹ Banks use business rules platforms to comply with the USA Patriot Act and anti-money-laundering laws. See the November 15, 2007, "[Business Rules And SOA Ease The Burden Of Regulatory Compliance](#)" report.
- ² Forrester identified BPM, business intelligence, and business rules platforms as the technical foundations for Dynamic Business Applications. See the September 24, 2007, "[The Dynamic Business Applications Imperative](#)" report.
- ³ Forrester describes one example of common rules to support multiple applications. See the November 15, 2007, "[Business Rules And SOA Ease The Burden Of Regulatory Compliance](#)" report.
- ⁴ Aside from Fair Isaac and CA, most of the vendors driving the business rules market have historically been small and specialized. Even ILOG and Pegasystems, which vie with Fair Isaac for leadership in rules revenue, each generate less than \$200 million in annual revenues. As the business rules market moves into the application development mainstream, this is changing. See the December 21, 2007, "[Trends: Business Rules Platforms 2008](#)" report.
- ⁵ Haley Limited is the new name for the company formerly known as RuleBurst. RuleBurst acquired Haley Systems during late 2007. It is continuing to develop both platforms that existed before the acquisition, as well as integrating its two platforms. See the December 21, 2007, "[Trends: Business Rules Platforms 2008](#)" report.
- ⁶ For more on SAP's acquisition of YASU Technologies, see the December 21, 2007, "[Trends: Business Rules Platforms 2008](#)" report.
- ⁷ Rete is the most well known inferencing algorithm used to process rules. Designed by Dr. Charles Forgy during the late 1970s, the Rete algorithm and several variations found their way into expert system inferencing engines during the 1980s. Several of the business rules platform vendors employ inferencing algorithms based upon the original Rete algorithm.
- ⁸ Red Hat JBoss Rules is descended from the Drools open source project, which has been operating for many years. See <http://www.redhat.com/jboss/frameworks/rules/>.
- ⁹ The combination of Haley Office Rules and Haley Determinations Engine was formerly known as RuleBurst 8.

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